



REPUBLIC OF TURKEY
HALAL ACCREDITATION AGENCY

HALAL ACCREDITATION CERTIFICATE

Laboratory Performing Halal Testing

**Mikrokim Laboratuvar Hizmeti Kimya Gıda
Danışmanlık Çevre Eğitim San. ve Tic. Ltd. Şti.**

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**Accredited in accordance with the OIC/SMIIC 35:2020
Standard for the attached scope.**

Halal Accreditation Number

2024-057

Halal Accreditation Scope

Shown in annex

Initial Halal Accreditation Date

07.06.2024

Revision Date

03.12.2025

Expiry Date

07.06.2029

This certificate is valid with the halal accreditation scope until the expiry date, as long as the laboratory fully complies with the OIC/SMIIC 35:2020 Standard, national legislation and Halal Accreditation Agency rules.

Zafer SOYLU

Chairman of the Board

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Testing Fields-Tested Materials/Products	Name of the Test	Testing Method (national, international standards, in-house methods)
Meat and Meat Products	Moisture Determination	AOAC 950.46
Animal Feed	Moisture Determination	29955 Sayılı Resmi Gazete (Yemin resmî kontrolü için numune alma ve analiz metotlarına dair yönetmelik)
Cheese	Moisture Determination	AOAC 948.12
Bread	Moisture Determination	TS 5000
Roasted, Ground Coffee	Moisture Determination	TS ISO 11294
Instant Coffee	Moisture Determination	TS ISO 3726
Food (Ash ContentCovers Foodswith Ash Amount in 100 Grams 0,07to 8 Grams.)	Determination of Ash	NMKL 173
Feed and Feed Additives	Determination of Ash	AOAC 942.05
Meat and Meat Products	pH Electrometric Method	TS 3136 ISO 2917
Instant Coffee	pH Electrometric Method	TS ISO 3726
Bread and Biscuit	Determination of Salt Titrimetric Method	TS 5000 TS 2383
Meat and Meat Products	Determination of Salt Titrimetric Method	AOAC 935.47
Meat and Meat Products	Determination of Starch Titrimetric Method	TS 6812
Food	Determination of Total Nitrogen and Protein Kjeldahl Method	NMKL No:6
Feed and Feed Additives	Determination of Total Nitrogen and Protein Kjeldahl Method	AOAC 2001.11
Food (Except Milk and Dairy Products)	Determination of Fat Content	NMKL 160

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Milk and Milk Products	Determination of Fats by Gerber Method	In-house Method- "MİKROKİM.SOP.02.165 Rev01"(Modified from Journal of AOAC International, Vol.84,No.5, 2001, s.1499-1507.)
Feed and Feed Additives	Determination of Crude Oil or Ether Extract	AOAC 920.39
Honey	Sugar Components (Glucose, Fructose, Saccharose, Lactose and Maltose)Analysis HPLC-RID Dedector	IHC (Part No: 7.2)
Chocolate	Sugar Components (Glucose, Fructose, Saccharose, Lactose and Maltose)Analysis HPLC-RID Dedector	AOAC 980.13
Food	Determination of MonosodiumGlutamate (MSG) (in Terms of L-GlutamicAcid) Spectrophotometric Method	In-house Method- "MİKROKİM.SOP.02.026 Rev04"(Modified from R-Biopharm L- GlutamicAcid Colorimetric Method)
Fish and Fish Products	Determination of Biogenic Amines (Histamine) HPLC - UV Method	NMKL 196
Food	Colour / Determination of Food Dyes (E - 102 Tartrazin / Acid Yellow 23, E - 104 Quinoline Yellow, E - 110 Sunset Yellow FCF, E - 120 Carminicacid / Acid Red 4, E - 122 Carmosin (Azorubin), E - 123 Amaranth / Acid Red 27, E - 124 Ponceau 4R / Acid Red 18, E - 127 Eritrosin B, E- 129 Allura Red AC, E -131 Patent Blue V / Acid Blue 3, E - 132 Indigotine / Indigo Carmine, E - 133 Brilliant Blue FCF / Acid Blue 9, E - 142 Green S / Acid Green 50) HPLC - UV / DAD Method	In-house Method- "MİKROKİM.SOP.02.177 Rev02"(Modified from NMKL 130, WatersApplication Note)
Food, Food Supplement	Determination of Acesulfame-K, Aspartame and Saccharin HPLC-UV /DAD Method	JAOAC Vol.76 No:2 TS EN 12856

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Food, Aroma and Food Reinforcement	Determination of Benzoic Acid and Sorbic Acid HPLC-UV/DAD Method	NMKL 124
Food	Determination of PHB (Methyl-Ethyl and Propyl P-Hydroxybenzoate) HPLC-UV/DAD Method	NMKL 124
Meat and Meat Products	Determination of Collagen and Hydroxyproline Spectrophotometric Method	AOAC 990.26 TS 6236 ISO 3496
Animal and Vegetable Oils, Extracted Oils	Determination of Fatty Acid Composition - Trans fats, Saturated fatty acids-Unsaturated fatty acids monounsaturated fatty acids-Polyunsaturated fatty acids (Butyric Acid (C4:0), Caproic acid (C6:0), Caprylic Acid (C8:0), Capric acid (C10:0), Undecanoic Acid (C11:0), Lauric Acid (C12:0), Tridecanoic Acid (C13:0), Myristic Acid (C14:0), Myristoleic Acid (C14:1), Pentadecanoic Acid (C15:0), Cis-10 - Pentadecanoic Acid (C15:1), Palmitic Acid (C16:0), Palmitoleic Acid (C16:1), Heptadecanoic Acid (C17:0), Heptadecenoic Acid (C17:1), Stearic Acid (C18:0), Elaidic Acid (C18:1n9t), Oleic acid (C18:1n9c), Linoleic acid (C18:2n6t), Linoleic acid (C18:2n6c), Arachidic Acid (C20:0), γ -Linolenic acid (C18:3n6), Gadoleic Acid (C20:1), Linolenic acid (C18:3n3), Heneicosanoic acid (C21:0), Cis-11,14-Eicosadienoic Acid (C20:2), Behenic Acid (C22:0), Cis-8,11,14 -Eicosatrienoic Acid (C20:3n6), Erucic Acid (C22:1n9), Cis-11,14,17 -Eicosatrienoic Acid (C20:3n3), Tricosanoic Acid (C23:0), Arachidonic Acid (C20:4n6), Cis-13,16 -Docosadienoic Acid (C22:2), Lignoceric Acid (C24:0), Cis-5,8,11,14,17 -Eicosapentaenoic Acid (C20:5n3), Nervonic Acid (C24:1) Cis-4,7,10,13,16,19-Docosahexaenoic Acid GC-FID Method	In-house Method- "MİKROKİM.SOP.02.179 Rev02" (Modified from Agilent Application Note, Analysis of Fatty Acids)



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Non-Alcoholic Beverages	Determination of Ethanol (Ethyl Alcohol) and Methanol (Methyl Alcohol) GC FID Method	In-house Method- "MİKROKİM.SOP.02.167 Rev04"(Modified from AOAC 983.13 and Alcohols Fast Analysis of Alcohols in Alcoholic Beverages Agilent App. Note)
Spice	Determination of Aflatoxin B ₁ and Total Aflatoxin (B ₁ +B ₂ +G ₁ +G ₂) HPLC-FLD Method	AOAC 999.07
Dried Fruits	Determination of Aflatoxin B ₁ and Total Aflatoxin (B ₁ +B ₂ +G ₁ +G ₂) HPLC-FLD Method	AOAC 999.07
Cereals and Cereal Products	Determination of Aflatoxin B ₁ and Total Aflatoxin (B ₁ +B ₂ +G ₁ +G ₂) HPLC-FLD Method	AOAC 991.31
Milk, Dairy Products, Milk-Based Baby Food	Determination of Aflatoxin M ₁ HPLC-FLD Method	TS EN ISO 14501 In-house Method- "MİKROKİM.SOP.02.160 Rev03"(Modified from R-Biopharm Milk Extraction Method for Aflatoxin M ₁ RefNo: A11-RP71 / 70N.V2)
Cereal Based Baby and Child Foods	Determination of Aflatoxin B ₁ and Total Aflatoxin (B ₁ +B ₂ +G ₁ +G ₂) HPLC-FLD Method	AOAC 2000.16
Oil Seeds and Products	Determination of Aflatoxin B ₁ and Total Aflatoxin (B ₁ +B ₂ +G ₁ +G ₂) HPLC-FLD Method	İşletme İçi Metot -" MİKROKİM. SOP.02.128 Rev No:4" (AOAC 991.31 standartdan modifiye edilmiştir.)
Feed (Mixed, Bovine, Poultry Feed, Cat-Dog Food)	Determination of Aflatoxin B ₁ and Total Aflatoxin (B ₁ +B ₂ +G ₁ +G ₂) HPLC-FLD Method	AOAC 2003.02
Cereals, Cereal Products, Cereal Based Baby and Child Supplementary Foods	Determination of Deoxynivalenol (DON) HPLC UV Method	In-house Method- "MİKROKİM.SOP.02.158 Rev04"(Modified from R-Biopharm Babyfood Deoxynivalenol extraction method Refno: A2 -P50.V8 / R-Biopharm Cereal Deoxynivalenol Extraction Method Refno: from A1 - P50.V10)
Feed	Determination of Deoxynivalenol (DON) HPLC UV Method	In-house Method- "MİKROKİM.SOP.02.066 Rev04"(Modified from R-Biopharm

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		Animal Feed Deoxynivalenol Extraction Method)
Corn and Corn Containing Products, Corn Containing Baby Foods	Determination of Fumonisin (FB ₁ +FB ₂) HPLC-FLD Method	In-house Method- "MİKROKİM.SOP.02.155 Rev04" (R-Biopharm-Fumonisin Extraction Method Ref. No: P31 / V9 / 16.07.12 AOAC2001.04
Chocolate and Cocoa	Determination of Ochratoxin A HPLC-FLD Method	R-Biopharm, Rhone Application Note Cacao Ochratoxin A Extraction Method
Spices	Determination of Ochratoxin A HPLC-FLD Method	In-house Method- "MİKROKİM.SOP.02.184 Rev01" R-biopharm Ochraprep Analysis of Spice A20 / P14 V.9
Coffee and Coffee Products	Determination of Ochratoxin A HPLC-FLD Method	In-house Method- "MİKROKİM.SOP.02.134 Rev00" (Modified from R-Biopharm Ochratoxin A App. Note Coffee, Ref Code: A1-P14.V9)
Cereal Based Baby and Kids Foods	Determination of Ochratoxin A HPLC-FLD Method	In-house Method- "MİKROKİM.SOP.02.082 Rev04" (R-Biopharm Ochratoxin A Method Baby Food)
Cereals and Cereal Products	Determination of Ochratoxin A HPLC-FLD Method	In-house Method- "MİKROKİM.SOP.02.133 Rev00" (Modified from R-Biopharm Rhone Ltd. Application Note Ochratoxin A Extraction Method Cereal P 14 Ref No: V16)
Feed	Determination of Ochratoxin A HPLC-FLD Method	In-house Method- "MİKROKİM.SOP.02.118 Rev04" (Modified from R-Biopharm Ochraprep Analysis of Animal Feed A18-P14.V7)
Fruit Juices and Purees Containing Apple, Fruit-Based Baby and Kids Foods	Determination of Patulin HPLC-UV Method	In-house Method- "MİKROKİM.SOP.02.104 Rev03" (Modified from R-Biopharm EAISIMIP Patulin P250 / P250B, Apple Puree)
Cereals, Cereal Products, Cereal Based Baby and	Determination of Zearalenon (ZON) HPLC-FLD Method	In-house Method- "MİKROKİM.SOP.02.157 Rev4" (Modified from R-Biopharm

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ChildSupplementary Foods		BabyfoodZearelenone extraction method Ref no:A4 -RP91.V1 R-Biopharm CerealZearelenone Extraction Method Ref: A4-RP91 RP90.V13)
Feed	Determination of Zearalenon (ZON) HPLC-FLD Method	In-house Method- "MİKROKİM.SOP.02.079 Rev03"(Modified from R-Biopharm inZearalenone Analysis of Animal FeedRP91 / RP90 / v15)
Food and Food Additivesi	Lead (Pb),Cadmium (Cd),Arsenic (As), Nickel (Ni),Crom (Cr),Cobalt(Co),Antimony (Sb), Calcium (Ca),Magnesium (Mg),Potassium(K), Sodium (Na),Mercury (Hg), Iron (Fe), Copper (Cu), Zinc(Zn), Manganese (Mn), Selenium (Se), Aluminium (Al), ICP MS Method	In-house Method- "MİKROKİM.SOP.02.238 Rev03" (NMKL186, Modified)
Food and Food Additivesi	Tin (Sn) Analysis ICP-MS Method	NMKL 191
Feed and Feed Additives	Lead (Pb),Cadmium (Cd),Arsenic (As), Nickel (Ni),Crom (Cr),Cobalt(Co),Antimony (Sb), Calcium (Ca),Magnesium (Mg),Potassium(K),Phosphorus (P), Sodium (Na),Mercury (Hg), Iron (Fe), Copper (Cu), Zinc(Zn), Manganese (Mn), Selenium (Se), Aluminium (Al) ICP MS Method	In-House Method- "MİKROKİM.SOP.02.236 Rev03" (NMKL186, Modified)
Feed and Feed Additives	Tin (Sn) Analysis ICP-MS Method	NMKL 191
Food Contanct Polymer ,Silicone,Paper/Board Based Food Packaging Materials and Materials	Determination of Structure FT-IR Method	In-House Method- "MİKROKİM.SOP.02.153 Rev05"(Modified from Polymer Data HandBook)
Plastic Materials and Materials to beused for Children, PolycarbonateType Plastics, Plastic BasedMaterials in Contact with Food	Specific migration [2,2-bis(4-hydroxyphenyl) propane (Bisphenol A)] Analysis HPLC-FLD Method	BS EN 14372

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Plastic Materials and Articles inContact with Foodstuffs	Total Migration Analysis in Evaporative Food Simulants Method 1-Full Immersion Method	TS EN 1186-1,TS EN 1186-3 Metot 1
Plastic Materials and Articles inContact with Foodstuffs	Total Migration Analysis in Evaporative Food Simulants Method 3-Bag Method	TS EN 1186-1,TS EN 1186-3 Metot 3
Plastic Materials and Articles inContact with Foodstuffs	Total Migration Analysis in Evaporative Food Simulants Method 4-Reverse Bag Method	TS EN 1186-1,TS EN 1186-3 Metot 4
Plastic Materials and Articles inContact with Foodstuffs	Total Migration Analysis in Evaporative Food Simulants Method 5-Filling Method	TS EN 1186-1,TS EN 1186-3 Metot 5
Plastic Materials and Articles inContact with Foodstuffs	Overall Migration Analysis High Temperature Substitution Method (FattyFoods) Modified Polyphenylene Oxide (MPPO)	TS EN 1186-1 TS EN 1186 - 13
Plastic Based Substances andMaterials in Contact with Food	Specific Migration Cadmium (Cd), Nickel (Ni), Cobalt (Co), Antimony (Sb), Iron (Fe), Copper(Cu), Zinc (Zn), Manganese (Mn), Aluminium (Al), Terbium (Tb), Europium(Eu), Barium (Ba), Lithium (Li), Gadalonium (Gd), Lantanium (La) ICP-MS Method	TS EN 13130-1
Dyes Used in Plastics Food ContactMaterias	Lead (Pb), Cadmium (Cd), Arsenic (As), Mercury (Hg), Antimony (Sb),Barium (Ba), Selenium (Se), Chromium (Cr) ICP-MS Method	NMKL 191 In-House Method- "MİKROKİM.SOP.02.236 Rev 02" (AP (89)1,NMKL 186, Modified)
Paper and Board and PlasticSubstances and Materials in Contactwith Food	Arsenic (As),Mercury (Hg),Kadmium (Cd),Lead (Pb) ICP-MS Method	TS EN 645, TS EN 647, In-House Method- "MİKROKİM.SOP.02.237 Rev01" (TS 12498Modified)
Porcelain, Ceramic and Glass BasedMaterials in Contact with Food	Determination of Lead (Pb) and Cadmium (Cd) Release ICP-MS Method	TGK 2012/30
Printed Paper and Board and Materials in Contact with Food	Determination of Chloride (Cl ₂) Titrimetric Method	Mohr Method TS EN 645, TS EN 647
Plastic Based Substances and	Determination of Formaldehyde Spectrophotometric Method	EUR 24815, TS CEN/TS 13130 -23

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Materials in Contact with Food		
Paper Based Packaging	Determination of Formaldehyde Spectrophotometric Method	EN 645/ EN 647 TS EN 1541
Food	Enumeration of Coliform Bacteria, Fecal Coliform and Escherichia coli MPN Technique - Colony Count Technique	FDA-BAM Chapter 4
Bread Yeast	Enumeration of Spores of B.subtilis	TS 3522
Food Feed Environmental Samples from Food Premises	Detection of <i>Salmonella</i> spp.	TS EN ISO 6579-1
Milk and Milk Products, Powdered Baby Food	Detection of <i>Cronobacter</i> spp.	ISO 22964
Food	Enumeration of Sulfite-Reducing Anaerobic Bacteria Colony Count Technique	ISO 15213-1
Food	Detection of <i>Campylobacter</i> spp.	ISO 10272-1
Food Feed	Enumeration of <i>Escherichia coli</i> Colony Count Technique	ISO 16649-2
Food Feed Environmental Samples from Food Premises	Detection of <i>Listeria monocytogenes</i>	ISO 11290-1
Food Feed Environmental Samples from Food Premises	Aerobic Plate Count Pour Plate Technique	ISO 4833-1
Food	Enumeration of <i>Bacillus cereus</i> Colony Count Technique	ISO 7932
Food	Enumeration of <i>Clostridium perfringens</i> Colony Count Technique	ISO 15213-2
Food	Detection of <i>Escherichia coli</i> O157	ISO 16654
Food Feed Environmental Samples from Food Premises	Enumeration of Enterobacteriaceae Colony Count Technique	ISO 21528-2

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Food Feed Environmental Samples from Food Premises	Enumeration of Coliform Bacteria Colony Count Technique	ISO 4832
Food Feed Environmental Samples from Food Premises	Enumeration of Yeast and Mold Colony Count Technique	ISO 21527-1 ISO 21527-2
Sea Food	Detection of <i>Vibrio cholerae</i> and <i>Vibrio parahaemolyticus</i>	ISO 21872-1
Food Feed Environmental Samples from Food Premises	Enumeration of Coagulase-Positive Staphylococci (<i>Staphylococcus aureus</i> and Other Species) Colony Count Technique	ISO 6888-1
Food Feed Environmental Samples from Food Premises	Enumeration of <i>Escherichia coli</i> MPN Technique	ISO 16649-3
Food Feed Environmental Samples from Food Premises	Detection of <i>Salmonella</i> spp.	AFNOR BRD 07/11-12/05 RAPID' Salmonella Agar-V10
Food Feed Environmental Samples from Food Premises	Detection of <i>Listeria monocytogenes</i>	AFNOR BRD 07/04-09/98 RAPID' L.mono-V11
Bread and Bread Types	Enumeration of Spores of <i>B.subtilis</i> MPN Technique	TS 5000
Food Feed Environmental Samples from Food Premises	Enumeration of Coagulase-Positive Staphylococci (<i>Staphylococcus aureus</i> and Other Species) MPN Technique	ISO 6888-3
Food Feed Environmental Samples from Food Premises	Enumeration of Yeast and Mold ($aw \leq 0,95$ and $aw > 0,95$) Colony Count Technique	AFNOR 01/13 – 07/14 RAPID Yeast and Mold Count Plate 3M Petrifilm
Food Feed Environmental Samples from Food Premises	Enumeration of Enterobacteriaceae MPN Technique	ISO 21528-1



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Food	Detection of Staphylococcal Enterotoxin ELISA Method	EU-RL European Screening Method Ver.5 (RIDASCREEN Set Total Kit R 4105)
Food Feed	GMO Screening Analysis (p35S /tNOS /pFMV) Real Time PCR Method Pretreatment: DNA Extraction	EURL Method (QL-ELE-00-012), EURL Method (QL-ELE-00-015), ISO21569, ISO 24276, Kit Method(Magnesia 16 Nucleic Acid Isolation KitManual), ISO 21571
Food	Determination of Meat Species ("Horse, Donkey" Meat Search) "Conventional PCR" Method Pretreatment: DNA Extraction	In-House Method "MİKROKİM.SOP.04.006 Rev03" (Modified from Kesmen, Z., Sahin, F., and Yetim, H., PCR assay for the identification of animal species in cooked sausages, Meat Sci, 77 (4), 649-653.) Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food	Determination of Meat Type ("Chicken, Turkey" Meat Search) "Conventional PCR" Method Pretreatment: DNA Extraction	In-House Method "MİKROKİM.SOP.04.008 Rev03" (I. Martin, T. Garcia, V.Fajardo, I. Lopez- Calleja, M. Rojas, M.A. Pavon et al. Technical Note: Detection of Chicken, Turkey, Duck and Goose Tissues in Feedstuffs using species- specific polymerase chain reaction, Journal of Animal Science, 85 (2007). pp. Modified from 452-458) Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food Feed	Species Identification (Beef DNA Search) Real Time PCR Method Pre-Process:DNA Extraction	Kit Method (Exon Screen Beef Assay) , ISO 21569, ISO 24276 Kit Method (Exon Screen Pork Assay), ISO 21569, ISO 24276
Food Feed	Tür Tespiti (Domuz DNA'sı Aranması) Real Time PCR Metodu Ön işlem: DNA Ekstraksiyonu	Kit Method (Exon Screen Pork Assay), ISO 21569, ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Gelatin and Gelatin Containing Foods	Determination of Meat Species ("Pork, Beef," Fish Meat Search) "Conventional PCR" Method Pretreatment: DNA Extraction	İşletme İçi Metot- "MİKROKİM.SOP.04.021 Rev03" (H. Cai., , X. Gu., M. S. Scanlan., D. H. Ramatlapeng., C. R. Lively. Real-time PCR Assays for Detection and

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		Quantitation of Porcine and Bovine DNA in Gelatin Mixtures and Gelatin Capsules- Sun, M., Liang, C., Gao, H., Lin, C., & Deng, M. (2009). Detection of Parvalbumin, a Common Fish Allergen Gene in Food, by Real-Time Polymerase Chain Reaction'den modifiye edilmiştir) Kit Metodu (Magnesia 16 Nükleik Asit İzolasyon Kit Manuel)
Meats and Foods Containing All Meat Products	Determination of Meat Species ("Duck" Meat Search) "Real Time PCR" Method Pretreatment: DNA Extraction	In-House Method "MİKROKİM.SOP.04.019 Rev 02" (N. Pegels., I. González., I. López-Calleja., S. Fernández., T. García. R. Martín. Evaluation of a TaqMan Real-Time PCR Assay for Detection of Chicken, Turkey, Duck, and Goose Material in Highly Processed Industrial Feed Samples, Departamento de Nutrición, Bromatología y Tecnología de los Alimentos Facultad de Veterinaria.) Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food	Determination of Milk Type ("Sheep, Goat and Cow" Milk Search) Real Time PCR Method Pretreatment: DNA Extraction	In-House Method "MİKROKİM.SOP.04.017 Rev02" (Restsch.J. et al. Modified from Eur Food Res Technol (2013) 236: 217-227) Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food Feed	Searching for Animal DNA (16S rRNA Gene) Conventional PCR Method Pretreatment: DNA Extraction	In-House Method - MİKROKİM.SOP.04.068 Rev 00 " (Food Control 43 (2014) 35-41.), ISO 21569 ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food Feed	Searching Plant Specific "Soybean" Gene Real Time PCR Method Pretreatment: DNA Extraction	EURL Method (QT-TAX-GM-009), ISO 21569, ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)

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Food Feed	Plant Specific Corn Gene Search Real Time PCR Method Pre-treatment: DNA Extraction	EURL Method (QT-TAX-ZM-011), ISO 21569, ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual, ISO 21571)
Food Feed	Plant Specific Canola (FatA) Gene Search Real Time PCR Method Pre-treatment: DNA Extraction	EURL METHOD (OT-TAX-BN-001), ISO 21569, ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food Feed	Plant Specific Cotton Gene Search Real Time PCR Method Pre-treatment: DNA Extraction	EURL METHOD (QT-TAX-GH-015), ISO 21569, ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food Feed	Plant Specific Rice Gene Search Real Time PCR Method Pre-treatment: DNA Extraction	EURL METHOD (QT-TAX-OS-017), ISO 21569, ISO 24276, Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food Feed	Plant Specific Potato Gene Search Real Time PCR Method Pre-treatment: DNA Extraction	EURL Method (QT-TAX-ST-010), ISO 21569, ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food Feed	Plant Specific Actin Gene Search Real Time PCR Method Pre-treatment: DNA Extraction	In-House Method (MİKROKİM.SOP.04.014 Rev02) " (I. Laube., H. Hird., P. Brodmann., S. Ullmann., M. Schöne-Michling. J. Chisholm. H. Broll. Development of Primer and Probe Sets for the Detection of Plant Species in Honey, Food Chemistry From 118 (2010) 979-986. Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food Feed	MON87701 Soy Type Determination Analysis Real Time PCR Method Pre-treatment: DNA Extraction	EURL METHOD (QT-EVE-GM-010), ISO 21569 ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food Feed	MON87708 Soy Type Determination Analysis Real Time PCR Method Pre-treatment: DNA Extraction	EURL METHOD (QT-EVE-GM-012), ISO 21569, ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food Feed	MON 87769 Soy Type Determination Analysis	EURL METHOD (QT-EVE-GM-002), ISO 21569,

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	Real Time PCR Method Pre-treatment: DNA Extraction	ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food Feed	CV127 (BPS-CV127-9) Soy Type Determination Analysis Real Time PCR Method Pre-treatment: DNA Extraction	EURL METHOD, (QT-EVE-GM-011), ISO 21569 ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food Feed	DP 305423-1 Soy Type Identification Analysis Real Time PCR Method Pre- treatment: DNA Extraction	EURL METHOD (QT-EVE-GM-008), ISO 21569 ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food Feed	DAS 44406-6 Soy Type Identification Analysis Real Time PCR Pretreatment: DNA Extraction	EURL METHOD, QT-EVE-GM-015, ISO 21569, ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual), ISO 21571
Food Feed	DAS 68416-4 Soy Type Identification Analysis Real Time PCR Pretreatment: DNA Extraction	EURL METHOD QT-EVE-GM-013, ISO 21569, ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual), ISO 21571
Food Feed	DAS 81419-2 Soy Type Identification Analysis Real Time PCR Pretreatment: DNA Extraction	EURL METHOD, QT-EVE-GM-014, ISO 21569, ISO 24276, Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual), ISO 21571
Food Feed	MON 87751-7 Soy Type Identification Analysis Real Time PCR Pretreatment: DNA Extraction	EURL METHOD QT-EVE-GM-016, ISO 21569, ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual), ISO 21571
Food Feed	DAS 40278-9 Corn Type Identification and Quantity Analysis Real Time PCR Method Pre-treatment: DNA Extraction	EURL METHOD (QT-EVE-ZM-004), ISO 21569, ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food Feed	73496 Canola Type Identification Analysis Real Time PCR Method Pretreatment: DNA Extraction	EURL METHOD (QT-EVE-BN-009), ISO 21569, ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual), ISO 21571
Food Feed	281-24-236 Cotton Type Determination Analysis Real Time PCR Method Pre- treatment: DNA Extraction	EURL METHOD (QT-EVE-GH-001a), ISO 21569, ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)



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Food Feed	3006-210-23 Cotton Type Determination Analysis Real Time PCR Method Pre-treatment: DNA Extraction	EURL METHOD (QT-EVE-GH-001b), ISO 21569 ISO 24276
Food Feed	GHB614 Cotton Type Determination Analysis Real Time PCR Method Pre- treatment: DNA Extraction	EURL METHOD (QT-EVE-GH-006), ISO 21569, ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food Feed	GHB 811 Cotton Type Determination Analysis Real Time PCR Method Pretreatment: DNA Extraction	EURL METHOD (QT-EVE-GH-013), ISO 21569, ISO 24276 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual), ISO 21571
Food	Gluten Determination Real Time PCR Method Pretreatment: DNA Extraction	In-House Method "MİKROKİM SOP.04.018 Rev02" (D. Zeltner, M. Glomb., D. Maede (2009). Real- Time PCR Systems for the Detection of the Gluten-Containing Cereals Wheat, Spelt, Kamut, Rye, Barley and Oat. Eur Food Res Technol Eur Food Res Technol (2009) 228: Modified from 321-330) Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Gluten-Free and Gluten-Reduced Whole Foods	Gluten Determination ELISA Method	Kit Method (R-Bipharm Ridascreen Gliadin / Art No: R7001)
Food	Plant Specific Chickpea Gene Screening Conventional PCR Method Pretreatment: DNA Extraction	In-House Method "MİKROKİM.SOP.04.020 Rev02" (R. Alary C. Buissonade., P. Joudrier., M., Gautier. Detection and Discrimination of Cereal and Leguminous Species in Chestnut Flour by Duplex PCR. Eur Food Res Technol (2007) 225: 427-434) Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food	Pistachio Allergen Gene Search Real Time PCR Method Pretreatment: DNA Extraction	In-House Method "MİKROKİM.SOP.04.016 Rev02" (B. Brezna et al. Modified from Eur Food Res Technol (2008) 228: 197-203.) Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)



ANNEX
HALAL ACCREDITATION SCOPE

Food	Searching for Almond Allergen Gene Real Time PCR Method Pretreatment: DNA Extraction	In-House Method "MİKROKİM.SOP.04.016 Rev02" (B. Brezna et al. Modified from Eur Food Res Technol (2008) 228: 197-203.) Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food	Walnut Allergen Gene Real Time PCR Method Pretreatment: DNA Extraction	In-House Method "MİKROKİM.SOP.04.016 Rev02" (B. Brezna et al. Modified from Eur Food Res Technol (2008) 228: 197-203.) Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food	Hazelnut Allergen Gene Search Real Time PCR Method Pretreatment: DNA Extraction	In-House Method "MİKROKİM.SOP.04.016 Rev02" (L. Píknova et al. Eur Food Res Technol (2008) Modified from 226: 1155-1158) Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food	Peanut Allergen Gene Search Real Time PCR Method Pretreatment: DNA Extraction	In-House Method "MİKROKİM.SOP.04.016 Rev02" (R. Köppel et al. Modified from Eur Food Res Technol (2010) 230: 367-374) Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food	Cashew Allergen Gene Search Real Time PCR Method Pretreatment: DNA Extraction	In-house Method- "MİKROKİM.SOP.04.016 Rev02" (A.Ehlert et al. Food Anal. Methods Modified from (2008) 1:136-143 Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food	Sesame Allergen Gene Search Real Time PCR Method Pretreatment: DNA Extraction	In-House Method "MİKROKİM.SOP.04.016 Rev02" (S. Mustorp et al. Modified from Eur Food Res Technol (2008) 226: 771-778.) Kit Method (Magnesia 16 Nucleic Acid Isolation Kit Manual)
Food	Allergen Detection (Searching for Milk Protein) ELISA Method Pretreatment: Protein Extraction	Kit Method (R-Biopharm Ridascreen Fast Milk/ Art No: R4652)
Food	Allergen Detection (Egg Protein Detection)	Kit Method (R-Biopharm Ridascreen Fast Ei/Egg Protein/Art No:R6402)

ANNEX
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	ELISA Method Pretreatment: Protein Extraction	
Water	Color Determination Spectrophotometric Method	SM 2120 C
Water	pH Determination Electrometric Method	TS EN ISO 10523
Water	Determination of Conductivity Electrometric Method	TS 9748 EN 27888
Water	Determination of Nitrite / Nitrite Nitrogen Spectrophotometric Method	TS 7526 EN 26777
Water	Determination of Ammonium / Ammonium Nitrogen Pretreatment: Distillation Method Measurement: Titrimetric Method	SM 4500 – NH3 B SM 4500 – NH3 C
Water	Determination of Chloride Titrimetric Method	TS 4164 ISO 9297
Water	Determination of Free Chlorine Spectrophotometric Method	TS EN ISO 7393-2
Water	<i>Escherichia coli</i> Count Membrane Filtration Technique	ISO 9308-1
Water	Coliform Bacteria Count Membrane Filtration Technique	ISO 9308-1
Water	Counting Fecal Enterococci Membrane Filtration Technique	TS EN ISO 7899-2
Water	<i>Pseudomonas aeruginosa</i> count Membrane Filtration Technique	TS EN ISO 16266
Water	Sulfite-Reducing Anaerobic Bacteria (Clostridia) Spores Search and Count Membrane Filtration Technique	TS 8020 EN 26461-2
Water	Colony Count (22 °C)	TS EN ISO 6222
Water	Colony Count (36 °C)	TS EN ISO 6222
Water	Detection and Enumeration of Pathogen Staphylococci Membrane Filtration Technique	NF T90-412





REPUBLIC OF TURKEY
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ANNEX
HALAL ACCREDITATION SCOPE

Water	<i>Clostridium perfringens</i> Count (Including Spores) Membrane Filtration Technique	98/83/EC Direktive
Water	Aluminium (Al), Arsenic (As), Copper (Cu), Mercury (Hg), Iron (Fe), Cadmium(Cd), Crom (Cr), Lead (Pb), Manganese (Mn), Nickel (Ni), Selenium (Se),Antimony (Sb), Sodium (Na), Calcium (Ca), Barium (Ba), Magnesium (Mg),Potasium (K), Boron (B) ICP MS Method	EPA 3015 A EPA 6020 B